

一、缺失 **groundtruth** 的样本

1.1 top-level sample 完全缺少 **groundtruth** (70)

分布：stage1 67 个，stage4 2 个，stage5 1 个。

```
stage1/1-email-163email-3
stage1/1-email-163email-4
stage1/1-email-gmail-5
stage1/1-email-gmail-6
stage1/1-email-Outlook-1
stage1/1-email-Outlook-2
stage1/1-email-Outlook-5
stage1/1-email-QQemail-3
stage1/1-email-QQemail-6
stage1/1-Tech community-CSDN
stage1/2-ai-ChatGPT-3
stage1/2-ai-ChatGPT-4
stage1/2-ai-DeepSeek-1
stage1/2-ai-DeepSeek-3
stage1/2-ai-DeepSeek-5
stage1/2-ai-Gemini-1
stage1/2-ai-Gemini-2
stage1/2-ai-gpt cherystudio-1
stage1/3-Messaging-dingding-3
stage1/3-Messaging-QQ-3
stage1/3-Messaging-QQ-5
stage1/3-Messaging-TIM-2
stage1/3-Messaging-TIM-3
stage1/3-Messaging-TIM-4
stage1/3-Messaging-TIM-5
stage1/3-Messaging-Wechat-2
stage1/3-Messaging-Wechat-3
stage1/4-Drive-baidu-1
stage1/4-Drive-baidu-4
stage1/4-Drive-Google Drive-2
stage1/4-Drive-Google Drive-3
stage1/4-Drive-Google Drive-4
stage1/4-Drive-Google Drive-5
stage1/4-Drive-kuake-2
stage1/4-Drive-OneDrive-1
stage1/4-Drive-OneDrive-4
stage1/5-meeting-Tencent-3
stage1/5-meeting-Tencent-5
stage1/5-meeting-Tencent-6
stage1/5-meeting-Zoom-5
stage1/5-Microsoft Teams-2
stage1/5-Microsoft Teams-3
```

```
stage1/6-workplace-feishu-1
stage1/6-workplace-feishu-3
stage1/6-workplace-GoogleWorkplace-2
stage1/6-workplace-GoogleWorkplace-3
stage1/6-workplace-Lark-2
stage1/6-workplace-Lark-4
stage1/6-workplace-youdao-3
stage1/6-workplace-youdao-4
stage1/7-Tech community-bokeyuan-2
stage1/7-Tech community-bokeyuan-3
stage1/7-Tech community-bokeyuan-4
stage1/7-Tech community-CSDN-2
stage1/7-Tech community-Medium-1
stage1/7-Tech community-Medium-3
stage1/7-Tech community-Medium-4
stage1/7-Tech community-Medium-5
stage1/7-Tech community-stackoverflow-1
stage1/8-git-GitCode-2
stage1/8-git-GitCode-4
stage1/8-git-Gitee-1
stage1/8-git-Gitee-4
stage1/8-git-GitHub-2
stage1/8-git-Github-4
stage1/8-git-Github-5
stage1/8-git-GitLab-1
stage4/e2e-10
stage4/e2e-5
stage5/Annotation
```

1.2 top-level sample 只有嵌套 session 有 **groundtruth** (8)

这类样本不是完全没标注，但顶层结构不完整，后续脚本如果只按 sample 根目录处理，很容易漏掉。

```
stage2/1-transport-copy-3
stage2/2-filestruct-pdfsplit-3
stage2/3-content-dataexport-3
stage4/e2e-1
stage4/e2e-3
stage4/e2e-8
stage4/e2e-9
stage5/Baidu
```

二、**groundtruth** 文件本身有错

2.1 JSON 解析失败（13）

case	文件	错误
stage2/1-transfer-print-2	groundtruth.json	空文件， Expecting value: line 1 column 1
stage2/1-transport-copy- 3/session_20260424_203808	groundtruth.json	Expecting , delimiter: line 3 column 29
stage2/2-filestruct-pdfsplit- 3/session_20260424_224052	groundtruth.json	Expecting , delimiter: line 8 column 32
stage2/2-filestruct-rename-2	groundtruth.json	Expecting , delimiter: line 2 column 18
stage2/3-content-dataexport- 3/session_20260424_212521	groundtruth.json	Expecting , delimiter: line 8 column 32
stage2/3-content-OCR-2	groundtruth.json	Expecting , delimiter: line 9 column 22
stage2/5-screen-screenshot-2	groundtruth.json	Expecting , ,

case	文件	错误
		delimiter: line 7 column 27
stage4/4-1	groungtruth.json	Expecting value: line 5 column 19
stage4/4-2	groundtruth.json	Expecting , delimiter: line 7 column 27
stage4/9-1	groungtruth.json	Expecting value: line 13 column 30
stage4/9-2	groundtruth.json	Expecting , delimiter: line 7 column 27
stage5/Baidu/session_20260424_221706	groundtruth.json	Expecting , delimiter: line 8 column 32
stage5/U2-Annotation-2	groundtruth.json	Expecting , delimiter: line 11 column 5

2.2 groundtruth 实际写成了事件日志 (2)

这两个 case 的 `groundtruth.json` 不是标注结构，而是更像 monitor 事件数组。

```
stage2/2-filestruct-rename-4  
stage2/3-content-copy-4
```

2.3 `groundtruth` 文件名拼写错误 (5)

```
stage2/5-screen-screenshot-1 -> grongtruth.json  
stage4/4-1 -> grongtruth.json  
stage4/9-1 -> grongtruth.json  
stage4/e2e-2 -> groudtruth.json  
stage4/e2e-7 -> groudtruth.json
```

2.4 `groundtruth` 被错误放进 `logs/` (6)

```
stage2/1-transfer-USB flash drive-1 -> logs/groundtruth.json  
stage2/2-filestruct-zip-1 -> logs/groundtruth.json  
stage2/3-content-copy-2 -> logs/groundtruth.json  
stage2/4-contentchange-base64-1 -> logs/groundtruth.json  
stage4/2-1 -> logs/groundtruth.json  
stage4/7-1 -> logs/groundtruth.json
```

三、附属文件缺失或结构不完整

3.1 缺少 `logs.json` (13)

```
stage1/0-normal-drive-dropbox-1  
stage1/0-normal-email-protonmail-1  
stage1/0-normal-git-gitee-1  
stage1/0-normal-workplace-googleworkspace-1  
stage1/2-ai-poe-1  
stage1/3-Messaging-Discord-1  
stage1/5-meeting-dingding-1  
stage1/7-Tech community-juejin-1  
stage2/4-contentchange-base64-4  
stage4/e2e-2  
stage4/e2e-7
```

```
stage5/U1-StegoImage-2
stage5/U5-ali-2
```

3.2 缺少 **keyevents.json** (3)

```
stage2/5-screen-record-3
stage4/e2e-2
stage4/e2e-7
```

3.3 缺少 **INDEX.md** (3)

```
stage4/11-1
stage4/e2e-2
stage4/e2e-7
```

3.4 缺少 **logs/** 与 **video/** 目录 (2)

这两个样本属于明显不完整的空壳样本。

```
stage4/e2e-2
stage4/e2e-7
```

四、标注内容与目录信息不一致

4.1 风险操作缺少 **sensitive_file_path** (1)

stage5/U4-Bluetooth-SensitiveLeak-2 的第 4 个风险操作 **operation** 已经描述了泄漏动作，但 **sensitive_file_path** 为空。

4.2 **record_id** 与目录名不一致 (62)

其中 55 个属于明显错配，已经不是简单大小写问题。

明显错配 (55)

```
stage1/0-normal-git-bitbucket-1 -> 0-normal-community-csdn-1
stage1/1-email-163email-2 -> 1
stage1/1-email-gmail-1 -> 2-ai-Gemini-3
stage1/1-email-Outlook-4 -> 1
stage1/1-email-QQemail-4 -> 1
stage1/2-ai-DeepSeek-4 -> 1
stage1/2-ai-Gemini-3 -> 3-Messaging-dingding-1
stage1/2-ai-gpt cherystudio-3 -> 1
stage1/3-Messaging-dingding-1 -> 4-Drive-baidu-2
stage1/3-Messaging-Wechat-1 -> 1
stage1/4-Drive-baidu-2 -> 4-Drive-OneDrive-2
stage1/4-Drive-Google Drive-1 -> 1
stage1/4-Drive-kuake-4 -> 1
stage1/4-Drive-OneDrive-2 -> 5-meeting-Zoom-1
stage1/5-meeting-Zoom-1 -> 6-workspace-youdao-1
stage1/5-meeting-Zoom-4 -> 1
stage1/5-Microsoft Teams-1 -> 1
stage1/5-Microsoft Teams-5 -> 5-meeting-Microsoft Teams-5
stage1/6-workplace-feishu-4 -> 1
stage1/6-workplace-GoogleWorkplace-1 -> 1
stage1/6-workplace-Lark-1 -> 6-workspace-Lark-1
stage1/6-workplace-Lark-5 -> 1
stage1/6-workplace-youdao-1 -> 7-Tech community-stackoverflow-2
stage1/6-workplace-youdao-2 -> 6-workspace-youdao-2
stage1/7-Tech community-CSDN-1 -> 1
stage1/7-Tech community-CSDN-4 -> 7-Tech community-CSDN-3
stage1/7-Tech community-Medium-2 -> 1
stage1/7-Tech community-stackoverflow-2 -> 8-git-Gitee-2
stage1/8-git-GitCode-1 -> 1
stage1/8-git-Gitee-2 -> 8-git-GitLab-2
stage1/8-git-Gitee-5 -> 1
stage2/1-transfer-print-1 -> 45
stage2/1-transfer-USB flash drive-1 -> 20260322-usb-transfer-tomato-4
stage2/2-filestruct-rename-1 -> 1-email-163email-2
stage2/2-filestruct-rename-3 -> 1
stage2/2-filestruct-zip-1 -> 20260322-leak-monitor-tomato-1
stage2/2-filestruct-zip-2/3-content-copy-1 -> 1
stage2/3-content-copy-2 -> session_20260319_194505
stage2/3-content-OCR-1 -> 1-email-163email-2
stage2/3-content-OCR-3 -> 1
stage2/4-contentchange-base64-1 -> 20260322-obfuscation-tomato-2
stage2/4-contentchange-transfer-1 -> 10
stage2/4-contentchange-transfer-2 -> 1-clipboard-email-1
stage2/5-screen-record-3 -> 1
stage2/5-screen-screenshot-1 -> 1-email-163email-2
stage4/2-1 -> 20260322-audit-tomato-005
stage4/3-1 -> 1
stage4/3-2 -> session_20260322_142439
stage4/7-1 -> 20260322-data-leak-audit-006
stage4/8-1 -> 10
stage4/8-2 -> session_20260322_143752
stage4/e2e-1/session_20260503_161617 -> e2e-1
stage4/e2e-1/session_20260505_213253 -> e2e-1
```

```
stage4/e2e-9/session_20260429_223203 -> 4-edit-wps-3
stage5/U4-Bluetooth-SensitiveLeak-2 -> U4-Bluetooth-SensitiveLeak-3
```

轻微不一致 (7)

这 7 个更像命名规范不统一，但仍建议修正。

```
stage1/2-ai-gpt cherystudio-2 -> 2-ai-gptcherystudio-2
stage1/2-ai-gpt cherystudio-5 -> 2-ai-gpt-cherystudio-5
stage1/4-Drive-baidu-5 -> 4-drive-baidu-5
stage1/7-Tech community-stackoverflow-3 -> 7-Tech community-StackOverflow-3
stage1/8-git-Gitee-3 -> 8-Git-Gitee-3
stage1/8-git-GitLab-4 -> 8-Git-GitLab-4
stage5/StegoImage -> Stegoimage
```

4.3 total_operations 与实际 operations 数量不一致 (3)

```
stage2/4-contentchange-base64-4 -> total_operations=2, actual_operations=3
stage4/1-1 -> total_operations=5, actual_operations=4
stage4/6-2 -> total_operations=2, actual_operations=3
```

五、keyevents.json 契约问题

根据 `tools/ScreenMonitor/windows_monitor/Log_Collection_Guide.md` 当前约定：

- `keyevents.json` 是最终用于标注的数据文件
- `app_switch` / `website_visit` 若存在，`file_path` 必须是完整精确路径
- 若无法精确绑定文件路径，这类事件不应该进入 `keyevents.json`

5.1 发现结果

- 共有 231 个 case 出现 `app_switch.file_path=""`
- 共涉及 1487 条 `app_switch` 事件
- 分布：

- **stage1**: 149 cases
- **stage2**: 46 cases
- **stage4**: 25 cases
- **stage5**: 11 cases

这不是个别脏样本，而是明显的系统性契约不一致。它不会一定导致 benchmark 无法运行，但会让“日志字段语义”与当前文档约定脱节，后续做规则、训练、可视化时都容易踩坑。

5.2 问题最集中的 case（前 15）

```
stage4/11-1 -> 116
stage4/11-2 -> 40
stage1/5-meeting-Tencent-1 -> 27
stage4/3-1 -> 25
stage4/6-2 -> 25
stage1/0-normal-community-csdn-1 -> 24
stage1/5-meeting-Zoom-3 -> 23
stage1/5-meeting-Zoom-2 -> 21
stage1/7-Tech community-CSDN-4 -> 18
stage1/1-email-fastmail-1 -> 17
stage1/5-meeting-Zoom-1 -> 17
stage1/0-normal-meeting-tencent-1 -> 16
stage4/9-1 -> 14
stage1/4-Drive-Google Drive-1 -> 13
stage1/5-meeting-Tencent-2 -> 13
```

5.3 其他空 **file_path** 事件

审计时还观察到其他 **file_path=""** 事件类型：

- **clipboard_text**: 519
- **clipboard_image**: 37
- **app_open**: 3
- **clipboard_operation**: 2

其中 **clipboard_*** 类事件允许无文件路径，因此这里不单独计为错误；本次文档重点记录明确违背契约的 **app_switch** 问题。